

$$Z = x_1 - x_2 \rightarrow \min$$

$$-Z = -x_1 + x_2 \rightarrow \max$$

$$\begin{cases} x_1 + x_2 \leq 10 \\ x_1 - 3x_2 \leq 3 \\ 3x_1 + x_2 \geq 9 \\ -x_1 + x_2 \leq 4 \\ x_1, x_2 \end{cases}$$

$\Rightarrow$

$$\begin{cases} x_1 + x_2 + x_3 = 10 \\ x_1 - 3x_2 + x_4 = 3 \\ 3x_1 + x_2 - x_5 = 9 \\ -x_1 + x_2 + x_6 = 4 \\ x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{cases}$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 10 \\ 1 & -3 & 0 & 1 & 0 & 0 & 3 \\ 3 & 1 & 0 & 0 & -1 & 0 & 9 \\ -1 & 1 & 0 & 0 & 0 & 1 & 4 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 10 \\ 0 & -4 & -1 & 1 & 0 & 0 & -7 \\ 0 & -2 & -3 & 0 & -1 & 0 & -21 \\ 0 & 2 & 1 & 0 & 0 & 1 & 14 \\ 0 & 2 & 1 & 0 & 0 & 0 & 10 \end{pmatrix}$$

$$-3 - \frac{1 \cdot 1}{1} = -4$$

$$0 - \frac{1 \cdot 1}{1} = -1$$

$$3 - 10 = -7$$

$$1 - \frac{1 \cdot 3}{1} = -2$$

$$0 - \frac{1 \cdot 3}{1} = -3$$

$$9 - 10 \cdot 3 = -21$$

$$1 - \frac{1 \cdot (-1)}{1} = 2$$

$$4 - 10 \cdot (-1) = 4 + 10 = 14$$

$$0 - \frac{1 \cdot (-1)}{1} = 1$$

$$1 - \frac{1 \cdot (-1)}{1} = 2$$

$$0 - \frac{1 \cdot (-1)}{1} = 1$$

$$0 - 10 \cdot (-1) = 10$$

$$\left(\begin{array}{cccc|c} 1 & 0 & \frac{3}{2} & \frac{1}{2} & 0 & 0 & \frac{33}{4} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & 0 & \frac{7}{4} \\ 0 & 1 & \frac{5}{2} & -\frac{1}{2} & -1 & 0 & -\frac{35}{4} \\ 0 & 0 & \frac{1}{2} & \frac{1}{10} & 0 & 1 & \frac{21}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{13}{2} \end{array} \right)$$

✓

$$1 - \left(\frac{1 \cdot 1 \cdot 1}{-4} \right) = \frac{3}{4}$$

$$0 - \frac{1 \cdot 1}{-4} = \frac{1}{4}$$

$$10 - \left(\frac{1 \cdot -7}{-4} \right) = 10 - \frac{7}{4} = \frac{40}{4} - \frac{7}{4} = \frac{33}{4}$$

$$-3 - \left(\frac{-1 \cdot -2}{-4} \right) = -3 + \frac{2}{4} = -3 + \frac{1}{2} =$$

$$= -\frac{6}{2} + \frac{1}{2} = -\frac{5}{2} \quad 0 - \left(\frac{1 \cdot -2}{-4} \right) = -\frac{1}{2}$$

$$-21 - \left(\frac{-1 \cdot -2 \cdot -7}{-4} \right) = -21 + \frac{14}{4} =$$

$$= -\frac{84}{4} + \frac{14}{4} = -\frac{70}{4} = -\frac{35}{2}$$

$$14 - \left(\frac{2 \cdot -7}{-4} \right) = 14 - \frac{14}{4} = \frac{56}{4} - \frac{14}{4} = \frac{42}{4} = \frac{21}{2}$$

$$10 - \left(\frac{1 \cdot 7 \cdot 2}{-4} \right) = 10 - \frac{14}{4} = \frac{40}{4} - \frac{14}{4} = \frac{26}{4} = \frac{13}{2}$$

$$1 - \left(\frac{1 \cdot 1 \cdot 2}{-4} \right) = 1 - \frac{2}{4} = \frac{1}{2}$$

$$0 - \frac{1 \cdot 2}{-4} = \frac{1}{2}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{10} & -\frac{3}{10} & 0 & 3 \\ 0 & 1 & 0 & -\frac{3}{10} & -\frac{1}{10} & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{5} & \frac{2}{5} & 0 & 7 \\ 0 & 0 & 0 & \frac{4}{10} & -\frac{1}{5} & 1 & 7 \\ 0 & 0 & 0 & \frac{4}{10} & -\frac{1}{5} & 0 & 3 \end{array} \right)$$

Исходные	нагрузка			x_4	x_5	x_6
x_1	x_2	x_3	b_i	a_{i4}	a_{i5}	a_{i6}
1	0	0	7	1	0	3
0	1	0	0	0	1	0
1	0	1	7	0	0	7
0	0	0	4	1	0	7
0	0	0	10	0	1	3

$$-Z = -Z(3; 0; 7; 0; 0; 7) = -3 + 0 = -3$$

Шаг 1:

И.П.		x_1	x_2	x_3	x_4	x_5	x_6	CO
x_1	3	1	0	0	$\frac{1}{10}$	$\frac{3}{10}$	0	30
x_2	0	0	1	0	$\frac{3}{10}$	$\frac{1}{10}$	0	-
x_3	7	0	0	1	$\frac{1}{5}$	$\frac{2}{5}$	0	35
x_4	7	0	0	0	$\frac{4}{10}$	$-\frac{1}{5}$	1	$\frac{35}{2}$
$-Z = -3$	3	0	0	0	$\frac{1}{10}$	$-\frac{1}{5}$	0	30

(КАНАЛИЗАЦИЯ ПРАВИЛЬНО, НЕ 30)

$$3 : \frac{1}{10} = 30$$

$$7 : \frac{3}{10} = \dots$$

$$\frac{70}{4} = 17.5$$

$$7 : \frac{1}{5} = 35$$

Шаг 2:

И.П.		x_1	x_2	x_3	x_4	x_5	x_6	CO
x_1	3	1	0	0	$\frac{1}{10}$	0	0	30

Таблица 2:

Б.с.		x_1	x_2	x_3	x_4	x_5	x_6	СД
x_1	$\frac{5}{4}$	1	0	0	0	$-\frac{1}{4}$	$-\frac{1}{4}$	—
x_2	$\frac{21}{4}$	0	1	0	0	$\frac{1}{4}$	$\frac{3}{4}$	—
x_3	$\frac{7}{2}$	0	0	1	0	$\frac{1}{2}$	$-\frac{1}{2}$	—
x_4	$\frac{35}{2}$	0	0	0	1	$\frac{1}{2}$	$\frac{5}{2}$	—
$-Z$	-4	0	0	0	0	0	-1	—

$$-Z = -Z\left(\frac{5}{4}; \frac{21}{4}; \frac{7}{2}; \frac{35}{2}; 0; 0\right) =$$

$$= -\frac{5}{4} + \frac{21}{4} = \frac{16}{4} = 4$$

$$3 - \frac{\frac{1}{10} \cdot 7}{\frac{4}{10}} = 3 - \frac{7}{10} \cdot \frac{10}{4} = 3 - \frac{7}{4} = \frac{12}{4} - \frac{7}{4} = \frac{5}{4}$$

$$0 - \left(-\frac{\frac{3}{10} \cdot 7}{\frac{4}{10}} \right) = \frac{21}{10} \cdot \frac{10}{4} = \frac{21}{4}$$

$$7 - \frac{\frac{1}{5} \cdot 7}{\frac{4}{10}} = 7 - \frac{7}{5} \cdot \frac{10}{4} = 7 - \frac{14}{4} = \frac{28}{4} - \frac{14}{4} = \frac{14}{4} = \frac{7}{2}$$

Транспортная оптимизация

$$-Z = -x_1 + x_2 \rightarrow \max$$

$$\begin{cases} x_1 + x_2 \leq 10 \\ x_1 - 3x_2 \leq 3 \\ 3x_1 + x_2 \geq 9 \\ -x_1 + x_2 \leq 4 \\ x_1, x_2 \geq 0 \end{cases}$$

1) $x_1 + x_2 = 10$

x_1	0	10
x_2	10	0

2) $x_1 - 3x_2 = 3$

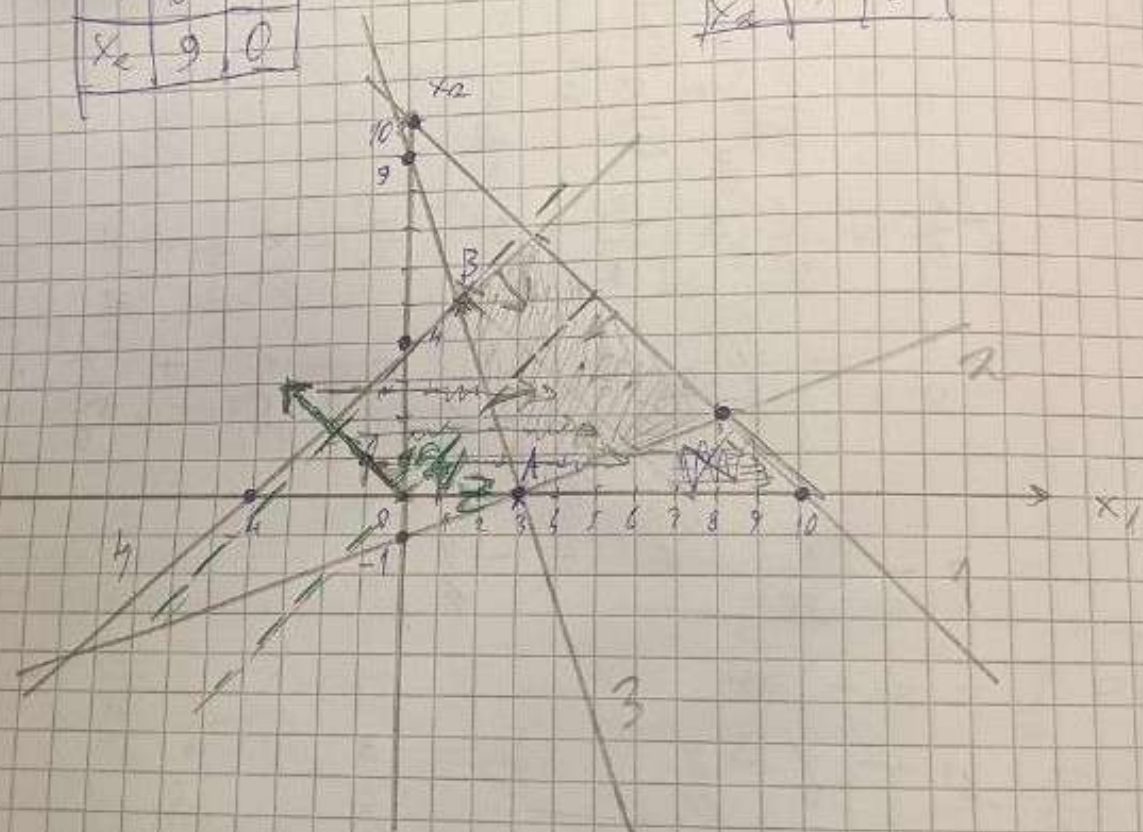
x_1	0	3
x_2	-1	0

3) $3x_1 + x_2 = 9$

x_1	0	3
x_2	9	0

4) $-x_1 + x_2 = 4$

x_1	0	-4
x_2	4	0



(0,0):

1) $0 + 0 \leq 10 \checkmark$

2) $0 - 3 \cdot 0 = 0 \leq 3 \checkmark$

3) $3 \cdot 0 + 0 = 0 \geq 9 \times$

4) $-0 + 0 = 0 \leq 4 \checkmark$

вариант	N	машина	мощность
3, 0, 7, 0, 0, 77	1		A
5, 20, 7, 35, 0, 0	2		B
4, 4, 1, 3, 2, 1, 0			

[Handwritten scribbles and notes]

$$\frac{5}{4} + \frac{29}{4} = \frac{34}{4} = 8,5 \leq 10 \text{ V}$$

$$\frac{5}{4} - 3 \cdot \frac{29}{4} = \frac{5 - 87}{4} = -\frac{82}{4} = -20,5 \leq 3 \text{ V}$$

$$3 \cdot \frac{5}{4} + \frac{29}{4} = \frac{15}{4} + \frac{29}{4} = \frac{44}{4} = 11 \geq 9 \text{ V}$$

$$-\frac{5}{4} + \frac{29}{4} = \frac{24}{4} = 6 \leq 4 \text{ V}$$

~~$$Z(18; 2) = 8 + 2 = 10$$~~

Ответ:

$$-Z_{\max} = -Z\left(\frac{5}{4}; \frac{29}{4}; \frac{9}{2}; \frac{35}{2}; 0; 0\right) = 4$$

$$Z_{\min} = -4$$

Однако это только одно решение

Так как мы в Z -списке таблицы 2 не
имеем R_1 в столбце базисной переменной
если 0, то по строке R_2 из левой,
используя оптимальное решение базиса

$$\begin{aligned} Z_{\min} &= Z\left(\frac{5}{4}R + 3(1-R); \frac{29}{4}R + 7(1-R)\right) = \\ &= Z\left(3 - \frac{7}{4}R; 7 - \frac{7}{4}R\right) = -4 \end{aligned}$$

~~$$Z(3; \sqrt{3}; \sqrt{3})$$~~

$$Z = 2X_1 + 3X_2 \rightarrow \max$$

$$\begin{cases} X_1 + X_2 \leq 5 \\ X_1 + 3X_2 \leq 9 \\ X_1 \leq 4 \\ X_1 + 2X_2 \leq 8 \\ X_1, X_2 \geq 0 \end{cases}$$

канон.

$$\begin{cases} X_1 + X_2 + X_3 = 5 \\ X_1 + 3X_2 + X_4 = 9 \\ X_1 + X_5 = 4 \\ X_1 + 2X_2 + X_6 = 8 \\ X_1, X_2, X_3, X_4, X_5, X_6 \geq 0 \end{cases}$$

$$\left(\begin{array}{cccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b \\ 1 & 1 & 1 & 0 & 0 & 0 & 5 \\ 1 & 3 & 0 & 1 & 0 & 0 & 9 \\ 1 & 0 & 0 & 0 & 1 & 0 & 4 \\ 1 & 2 & 0 & 0 & 0 & 1 & 8 \\ 2 & 3 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Iteration 1:

f.r		x_1	x_2	x_3	x_4	x_5	x_6	CO
x_1	1	1	1	1	0	0	0	5
x_3	5	1	1	1	0	0	0	3
x_4	9	1	3	0	1	0	0	-
x_5	4	1	0	0	0	1	0	-
x_6	8	1	2	0	0	0	1	4
$Z=37$	0	-2	-3	0	0	0	0	<u>37</u>

$Z = Z(0; 0, 5, 9, 4) = 2 \cdot 0 + 3 \cdot 5 = 10 + 27 = 37$

Iteration 2:

f.r		x_1	x_2	x_3	x_4	x_5	x_6	CO
x_3	2	$\frac{2}{3}$	0	1	$\frac{1}{3}$	0	0	3
x_2	3	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	0	3
x_5	4	1	0	0	0	1	0	4
x_6	2	$\frac{1}{3}$	0	0	$\frac{2}{3}$	0	1	6
$Z=9$	9	-1	0	0	1	0	0	<u>9</u>

$5 - \frac{1 \cdot 9}{3} = 5 - 3 = 2$; $1 - \frac{1 \cdot 1}{3} = \frac{2}{3}$

$4 - \frac{9 \cdot 0}{3} = 4$; $8 - \frac{0 \cdot 2}{3} = 8$

$0 - \frac{9 \cdot (-1)}{3} = 3$; $1 - \frac{1 \cdot 2}{3} = \frac{1}{3}$

$$Z = Z(0; 3; 2; 0; 4; 2) = 2 \cdot 0 + 3 \cdot 3 = 9$$

Iteration 3:

f	1	x_1	x_2	x_3	x_4	x_5	x_6	CD
x_1	3	1	0	$\frac{3}{2}$	$-\frac{1}{2}$	0	0	—
x_2	2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	—
x_5	1	0	0	$-\frac{3}{2}$	$\frac{1}{2}$	1	0	—
x_6	1	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	0	1	—
Z	12	0	0	$\frac{3}{2}$	$\frac{1}{2}$	0	0	2/3/12
		>0		>0				

$$0 - 1 \cdot \left(-\frac{1}{3}\right) = \frac{1}{3} \quad ; \quad -\frac{2}{3} - \left(\frac{1}{3} \cdot -\frac{1}{3}\right) = -\frac{2}{3} + \frac{1}{9} = -\frac{5}{9}$$

$$1 - \left(-1 \cdot -\frac{1}{3}\right) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$Z = Z(3; 2; 1; 1) = 2 \cdot 3 + 3 \cdot 2 = 6 + 6 = 12$$

Задача на максимизацию

$$Z = 2x_1 + 3x_2 \rightarrow \max$$

$$\begin{cases} x_1 + x_2 \leq 5 \\ x_1 + 3x_2 \leq 9 \\ x_1 \leq 4 \\ x_1 + 2x_2 \leq 8 \\ x_1, x_2 \geq 0 \end{cases}$$

$$1) \quad x_1 + x_2 = 5$$

x_1	0	5
x_2	5	0

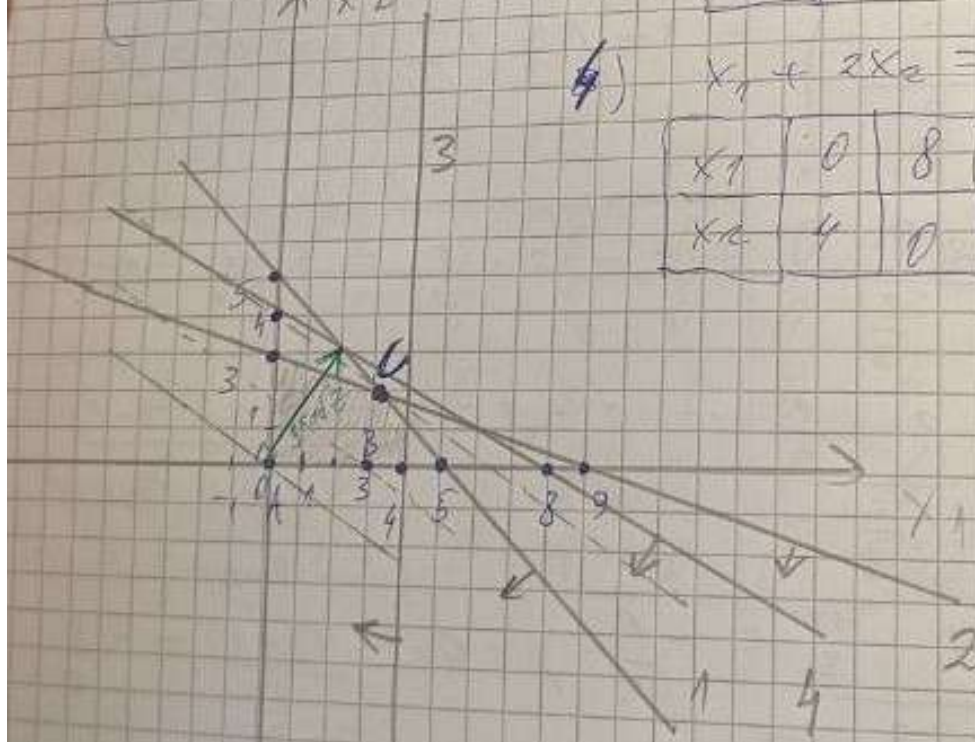
$$x_1 + 3x_2 \leq 9$$

$$2) \quad \begin{array}{|c|c|c|} \hline x_1 & 0 & 9 \\ \hline x_2 & 3 & 0 \\ \hline \end{array}$$

$$3) \quad x_1 \leq 4$$

$$4) \quad x_1 + 2x_2 = 8$$

x_1	0	8
x_2	4	0



$$(0, 0): \quad 1) \quad 0 + 0 = 0 \leq 5 \quad \checkmark$$

$$2) \quad 0 + 3 \cdot 0 = 0 \leq 9 \quad \checkmark$$

$$3) \quad (0, 0) \quad 0 \leq 4 \quad \checkmark$$

$$4) \quad 0 + 2 \cdot 0 = 0 \leq 8 \quad \checkmark$$

Resource	Madura	Moron
0, 0, 5, 3, 4	1	A
0, 3, 2, 0, 4, 2	2	B
5, 2, 0, 0, 1, 1	3	C

Answer: $Z_{\max} = Z(3, 2) = 12$

$$Z = 2x_1 - 8x_2 \rightarrow \min$$

$$\begin{cases} -x_1 + 4x_2 \leq 8 \\ 2x_1 - x_2 \leq 4 \\ x_1 + x_2 \geq 1 \\ x_1, x_2 \geq 0 \end{cases}$$

$$-Z = -2x_1 + 8x_2 \rightarrow \max$$

$$\begin{cases} -x_1 + 4x_2 + x_3 = 8 \\ 2x_1 - x_2 + x_4 = 4 \\ x_1 + x_2 - x_5 = 1 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

$$\left(\begin{array}{ccccc|c} -1 & 4 & 1 & 0 & 0 & 8 \\ 2 & -1 & 0 & 1 & 0 & 4 \\ 1 & 1 & 0 & 0 & -1 & 1 \\ -2 & 8 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccccc|c} 1 & -4 & -1 & 0 & 0 & -8 \\ 0 & 7 & 2 & 1 & 0 & 20 \\ 0 & 5 & 1 & 0 & -1 & 9 \\ 0 & 0 & -2 & 0 & 0 & -16 \end{array} \right)$$

$$-1 - \frac{4 \cdot 2}{-1} = -1 + 8 = 7; \quad 0 - \frac{1 \cdot 2}{-1} = 2;$$

$$4 - \frac{8 \cdot 2}{-1} = 4 + 16 = 20; \quad 1 - \frac{4 \cdot 1}{-1} = 5;$$

$$0 - \frac{1 \cdot 1}{-1} = 1; \quad 1 - \frac{8 \cdot 1}{-1} = 1 + 8 = 9$$

$$8 - \frac{4 \cdot (-2)}{-1} = 8 - 8 = 0; \quad 0 - \frac{1 \cdot (-2)}{-1} = -2$$

$$0 - \frac{8 \cdot (-2)}{-1} = -16$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & \frac{1}{7} & \frac{4}{7} & 0 & \frac{34}{7} \\ 0 & 1 & \frac{2}{7} & \frac{1}{7} & 0 & \frac{20}{7} \\ 0 & 0 & -\frac{3}{7} & -\frac{5}{7} & -1 & \frac{37}{7} \\ 0 & 0 & -2 & 0 & 0 & -16 \end{array} \right)$$

$$-1 - \frac{(-4 \cdot 2)}{7} = -\frac{7}{7} + \frac{8}{7} = \frac{1}{7}$$

$$0 - \frac{(1 \cdot -4)}{7} = \frac{4}{7}$$

$$-8 - \frac{(20 \cdot -4)}{7} = -\frac{8}{7} + \frac{80}{7} = \frac{72}{7}$$

		x_1	x_2	x_3	x_4	x_5	x_6
$\bar{z}$	1						
x_1	$\frac{24}{7}$	1	0	$\frac{1}{7}$	$\frac{4}{7}$	0	$\frac{24}{7}$
x_2	$\frac{28}{7}$	0	1	$\frac{2}{7}$	$\frac{1}{7}$	0	$\frac{20}{7}$
x_3	$-\frac{37}{7}$	0	0	$-\frac{3}{7}$	$-\frac{5}{7}$	-1	$-\frac{37}{7}$
$\bar{z}$	+16	0	0	+2	0	0	-16

optimal solution
found

		x_1	x_2	x_3	x_4	x_5	$\bar{z}$
$\bar{z}$	1						
x_1	$\frac{24}{7}$	1	0	$\frac{1}{7}$	$\frac{4}{7}$	0	$\frac{24}{7}$
x_2	$\frac{28}{7}$	0	1	$\frac{2}{7}$	$\frac{1}{7}$	0	$\frac{20}{7}$
x_3	$-\frac{37}{7}$	0	0	$\frac{3}{7}$	$\frac{5}{7}$	1	$-\frac{37}{7}$
$\bar{z}$	+16	0	0	+2	0	0	-16

row 0 = 0

no
more
improvement
possible

(optimal
solution)

$$-\bar{z} = -\bar{z} \left(\frac{24}{7}, \frac{20}{7}, 0, 0, \frac{37}{7} \right)$$

$$= -2 \cdot \frac{24}{7} + 8 \cdot \frac{20}{7} = -\frac{48}{7} + \frac{160}{7} = \frac{112}{7} = 16$$

$$Z_{min} = \bar{z} \left(\frac{24}{7}, \frac{20}{7}, 0, 0, \frac{37}{7} \right) = -16$$

feasible region is unbounded

$$-\bar{z} = -2x_1 + 8x_2 \rightarrow \max$$

$$\begin{cases} -x_1 + 4x_2 \leq 8 \\ 2x_1 - x_2 \leq 4 \\ x_1 - x_2 \geq 1 \\ x_1, x_2 \geq 0 \end{cases}$$

$$1) -x_1 + 4x_2 = 8$$

x_1	0	-8
x_2	2	0

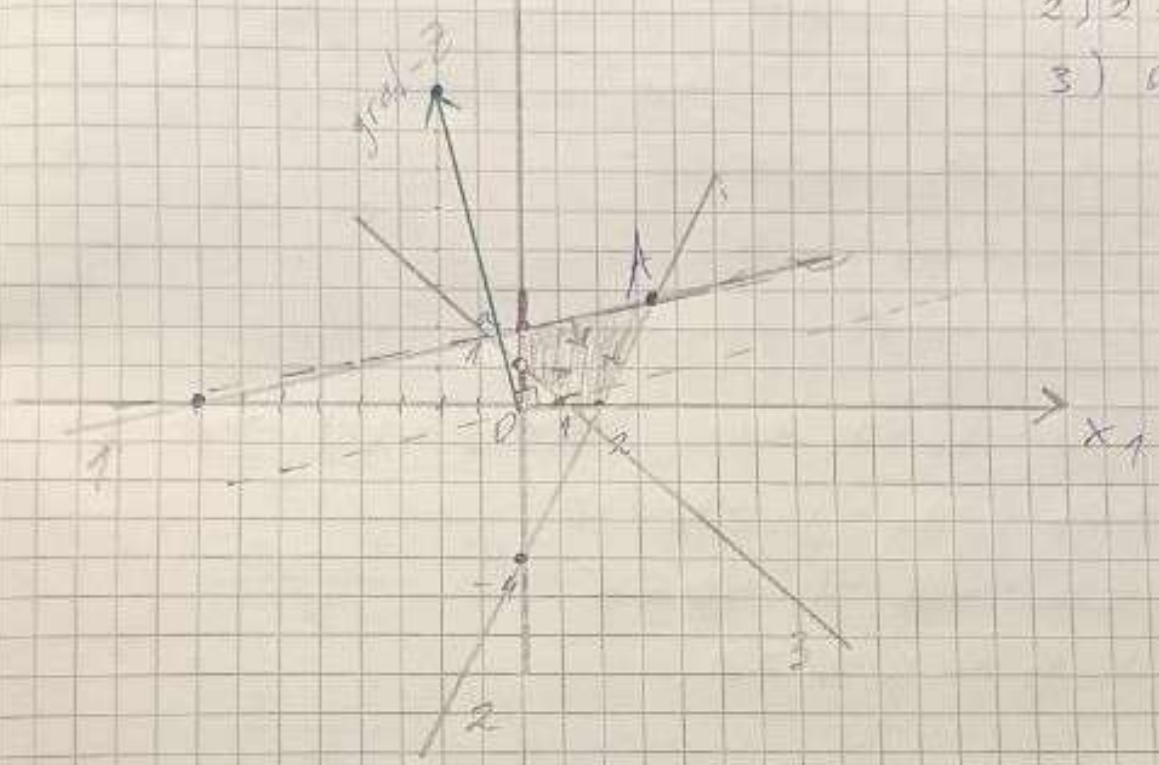
$$2) 2x_1 - x_2 = 4$$

x_1	0	2
x_2	-4	0

$$3) x_1 - x_2 = 1$$

x_1	0	1
x_2	1	0

$(0;0)$ 1) $-0 + 1 \cdot 0 = 0$
 2) $2 \cdot 0 - 0 = 0$
 3) $0 + 0 \geq 1$ X



Строки	Правильно	Правда
X $\frac{24}{7}, \frac{20}{7}, 0, 0, \frac{37}{7}$	1	A

Ответ:

$$Z_{min} = Z\left(\frac{24}{7}; \frac{20}{7}; 0; 0; \frac{37}{7}\right) = -16$$

Вопрос: это только одно решение?

Так как правильно в Z-строке таблицы 1

В строке базисной переменной есть 0, то
 то means 12 из которых,
 множество оптимальных решений бесконечно.

$$\begin{aligned}
 Z_{min} &= Z\left(\frac{24}{7}(1-\lambda); 2\lambda + \frac{20}{7}(1-\lambda)\right) = \\
 &= Z\left(\frac{24}{7} - \frac{24}{7}\lambda; \frac{20}{7} - \frac{4}{7}\lambda\right) = -16
 \end{aligned}$$